**Math 40a: Introduction to Applied Mathematics**
Homework 4

Upload your write-up and Python files to Moodle to submit your assignment. The write-up should be a self contained pdf—the graders should be able to evaluate your lab based on the write-up without having to look at your separate code files. If you choose to upload a pdf of an IPython notebook, it is okay if your code is included within the writeup, but please make sure to include clear answers to the questions.

## Problem 1 *(Predator Prey Models)*

The following is a simple model to describe the interactions between predator and prey populations:

$$\frac{dR}{dt} = \alpha R - \beta RF$$
$$\frac{dF}{dt} = -\gamma F + \delta RF.$$

where $R(t) \geq 0$ and $F(t) \geq 0$ denote the rabbit (prey) and fox (predator) populations, and $\alpha$, $\beta$, $\gamma$, and $\delta$ are positive parameters.

(a) What are the units of the parameters $\alpha$, $\beta$, $\gamma$, and $\delta$? Discuss their biological intepretations.

(b) Set $\alpha = 1.5$, $\beta = 0.5$, $\gamma = 0.4$, and $\delta = 0.8$. Modify your code from the lab and solve the system for initial populations of

   (a) 5 foxes and 0 rabbits,
   (b) 5 foxes and 10 rabbits,
   (c) 0 foxes and 10 rabbits.

   Plot the rabbit and fox populations for each of these cases, and interpret the results.

## Problem 2 *(Review of the original SIR Kermack–McKendrick paper)*

The original SIR paper by Kermack and McKendrick that was discussed in class is available on the course website. Please study the paper and answer the following questions. Note that we do not expect you to understand the paper in its entirety – one of the aims of this class is to glean the main points of a paper even if some of the details remain opaque.

(a) The paper makes a model of a flu epidemic in which the population is divided into three groups: the susceptible, the infected, and the recovered. Explain what each of these groups are.

(b) A few years ago there was a global fear about a bird flu pandemic. Imagine there was a single person in Boston who had bird flu and nobody else was exposed to the disease. Explain what the infected, susceptible, and recovered populations would be in this case.

(c) The paper makes several assumptions about how a flu epidemic progresses through a population. List two of these assumptions.

(d) Explain what is plotted in the figure on page 714.

(e) The central result of the paper is that it proposes a formula for a threshold population density above which a flu epidemic will spread and potentially become large. What does this threshold population density depend upon? *[Hint: See the discussion on page 715. You can ignore or skim the dense mathematics that occurs in the pages before this—the main point is on this page.]*

## Problem 3 *(High risk and low-risk populations)*

(a) First, consider the standard SIR model,

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

Suppose that $\beta = 0.1$, and $\gamma = 20$, and consider a population of one hundred people where initially $S = 90$, $I = 10$, and $R = 0$. Calculate the basic reproduction rate $R_0$. Solve the differential equation system numerically for $0 \leq t \leq 1.2$, and by looking at the final steady-state value of $R$, estimate the total number of people who become sick.

(b) Now suppose that the population in part (a) also comes into contact with a second, high-risk population. Let the number of susceptible, infected, and recovered people in this population be $S_H(t)$, $I_H(t)$, and $R_H(t)$ respectively. A combined model for this is given by the system of differential equations

$$\begin{aligned}\frac{dS}{dt} &= -S(\beta I + \beta_2 I_H) \\ \frac{dI}{dt} &= S(\beta I + \beta_2 I_H) - \gamma I \\ \frac{dR}{dt} &= \gamma I \\ \frac{dS_H}{dt} &= -S_H(\beta_2 I + \beta_3 I_H) \\ \frac{dI_H}{dt} &= S_H(\beta_2 I + \beta_3 I_H) - \gamma I_H \\ \frac{dR_H}{dt} &= \gamma I_H\end{aligned}$$

If $\beta_2 = 0$, then this system represents two independent SIR populations. However, if $\beta_2 > 0$ the two populations are coupled, and infected people in the high-risk population can infect people in the original population and vice versa. Solve the model numerically for the parameter values of $S = 90$, $I = 10$, $R = 0$, $S_H = 100$, $I_H = 0$, $R_H = 0$, $\beta = 0.1$, $\beta_2 = 0.05$, $\beta_3 = 0.5$, and $\gamma = 20$ over the the time interval $0 \leq t \leq 1.2$. Using your results to estimate the total number of people who become sick in both the original and high-risk populations.

(c) Describe a real-life example where the situation considered in part (b) would be a good model. For your example, discuss how the model may influence public policy decisions, such as vaccination strategies.